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Comps - B

- **Company:** Dolat Capital
- **Job Profile:** Quant Developer
- **Offer Type:** Internship + Placement
- **CTC:** ₹11 LPA (₹10 LPA fixed + ₹1 LPA variable)
- **Location:** Mumbai
- **Process Date:** September 2, 2025

Placement Process

This company is very approachable if you have a strong command of **C++**.

After not receiving a PPO from my internship at Barclays, I hadn't prepared much for the final placements as I had been fully focused on my internship. I consider myself lucky because Dolat Capital had already visited the campus once while I was interning, but they returned for a second round of hiring. This time, students who had applied previously were not eligible, which created a new opportunity for people like me.

Online Assessment (OA)

For the OA, only 20 students were shortlisted, mostly from MCA/M.Tech programs and those who hadn't applied during the company's first visit. The assessment consisted of five sections.

1. **C++ MCQs:** The questions were quite tricky. Some were based on concepts like the `static` and `volatile` keywords.

   one question was like

   class Person{

   string name;

   Person(string name){

   name=name;   //what will happened here

   }

   };

Int main(){

Person p1(“sumit”);

Printf(“my name is %s”, p1.name);// so basically when constructor is called without this

//As compiler in person class do not know about which name is used it just assigned its value to itself //and does not touches the class data member basically in constructor this was required;

}

2. **Python MCQs:** I had never coded much in Python, but the questions were manageable if you understood basic syntax like `for` loops and array slicing (e.g., `arr[:-1]`).
3. **Python Coding Question:** The task was to remove outliers from a given array. An outlier was defined as any element greater than the upper bound or less than the lower bound, where:
   - `upper = mean + standard_deviation`
   - `lower = mean - standard_deviation` I used NumPy's standard functions, `np.mean()` and `np.std()`, to solve this. There seemed to be a compiler issue on the platform; the same code that compiled once failed on a subsequent submission. I made sure to submit my working code and not touch it again, and I reported the incident to the placement cell.
4. **C++ Coding Question:** I was asked to design a parking system, which I felt was the most difficult question among all the candidates. The pricing logic was as follows:
   - If 1 vehicle is in the parking lot, the charge is ₹100 per car.
   - If 2 vehicles are present, the charge is ₹200 per car.
   - If 3 or more vehicles are present, the charge is ₹300 per car.

   The input consisted of a car's name, start time, and end time. You had to implement a `Parking` class to calculate the total cost for each car.

   Input:

   ```
   A 10 20
   B 10 20
   A 15 20
   A 30 40
   ```

   Output:

   ```
   Cost for A is 400  // (100 for 10-15) + (200 for 15-20) + (100 for 30-40)
   Cost for B is 100
   ```

   As a competitive programmer (subtle flex, lol), this was quite easy for me. While it could have been solved with a brute-force approach, I had seen this problem before and implemented an optimal solution using a **sweep line algorithm**.

5. **Aptitude:** The questions were a mix of easy and tough problems.

💡 **OA Tip:** To clear the assessment, focus on the Python and C++ coding questions. The test cases seemed limited, so even a brute-force solution could potentially pass.

After the OA, 5 students were shortlisted for interviews.

Interview Round 1 (In-Office)

This round was held at their office and covered technical and quantitative topics.

Technical & DSA Questions

1. **What is abstraction?** I was asked to explain the concept and provide a code example, detailing how virtual functions work internally.
2. **Difference between `struct` and `class` in** C++

    ```
    struct Node {
        int data;
        Node* next;
    };
    ```

    Then, I explained the difference between a `struct` and a `class` in C++. Many people confuse C structs with C++ structs. I knew the key difference is that members of a `struct` are **public** by default, whereas members of a `class` are **private** by default. The interviewer seemed impressed, and at that moment, I felt confident I would clear the technical round.

3. **Linked List Reordering:** I was asked a common LeetCode problem: reorder a linked list to connect the i-th node to the (n-i)-th node.
    - **Input:** `1 -> 2 -> 3 -> 4`
    - **Output:** `1 -> 4 -> 2 -> 3` I didn't need to pretend to think about it. I immediately gave the optimal solution: find the middle of the list using a **slow and fast pointer**, reverse the second half, and then merge the two halves. I wrote the full code on paper, which took almost two pages, ensuring it was clean and modular with separate functions for each step.

Probability & Statistics Questions

The next interviewer started with quantitative questions.

1. **Expected value of a fair six-sided die?** (Answer: 3.5)
2. **Expected number of rolls to get a 6?** (Answer: 6)
3. He then asked me to write a **recurrence relation** for the expected number of rolls to get a 5 followed immediately by a 6. I wasn't sure how to formulate the exact recurrence relation, so I wrote down the basic recursion logic, and he was satisfied with that.
4. **Distributions:** I was asked to name some common distributions. I mentioned **Binomial**, **Normal**, and **Uniform** distributions. He followed up by asking for the PDF (Probability Density Function) and CDF (Cumulative Distribution Function) of each, which I knew.
5. **Sum of Normal Distributions:** The final question was how to derive the distribution of $N_3=N_1+N_2$, where N_1 and N_2 are independent normal distributions. I didn't know the exact proof but tried to formulate an approach using variable transformations to show that I don't give up easily.

HR Interview

The HR interview was more of a formality. The questions were standard:

- Tell me about yourself.
- What is your CGPA?
- Tell me about your family background.
- What are 3 of your greatest strengths and 3 of your weaknesses?

Interview Round 3 (Online C++ Round)(confirmatory round which was online)

Two students were shortlisted for this final online round, which took place the next day.

1. **Implement a Data Structure:** I was asked to implement a class with the following functionalities:
   - `add(a)`: Adds a number to a stream.
   - `avg()`: Returns the average of the last `k` numbers added. I immediately proposed an optimized approach using a **queue**. We can push elements into the queue. If the queue size exceeds `k`, we pop from the front. This maintains a window of the last `k` elements.

   He then added a follow-up:

   - `update(new_k)`: The value of `k` can be changed. For this, I suggested using a `std::vector` and a **prefix sum array**. With a prefix sum array, we can find the sum of the last `k` elements in O(1) time. He then asked if it was possible to implement it without the extra space of the prefix sum array. I thought about it for a bit, and he said it was okay and moved to the next question.
2. **Delete a Node in a Doubly Linked List:** This was a straightforward question. First, he asked me to write the `struct` for a doubly linked list node.

   C++

   ```cpp
   struct Node {
       int data;
       Node* next;
       Node* prev;
   };
   ```

   Then, he asked me to write a function to delete a node given only a pointer to that node. This is my implementation:

   C++

   ```cpp
   void deleteNode(Node* ptr) {
       if (ptr == nullptr) {
           return;
       }

       // If the node to be deleted is not the head
       if (ptr->prev != nullptr) {
           ptr->prev->next = ptr->next;
       }
   ```

```
    // If the node to be deleted is not the tail
    if (ptr->next != nullptr) {
        ptr->next->prev = ptr->prev;
    }

    // Free the memory to avoid a memory leak
    delete ptr;
}
```

After I wrote the code, he asked me what a **memory leak** is. I was prepared for this and explained it thoroughly. This is a very common interview question, so it's good to practice it. My code was exactly what he was looking for, and I was confident I would be selected.

In the end, I was the only one selected from this round. Previously, they had hired two others, bringing the total to three students.

Useful Tips

This company is a great fit for you if you know C++ well. In addition to DSA, you should focus on advanced C++ concepts:

- **OOPs principles** (especially the internal working of `virtual` functions).
- Templates
- Multithreading (e.g., spinlocks)
- Smart Pointers (`unique_ptr`, `shared_ptr`)
- Implementing your own `string` and `vector` classes.

For these topics, I highly recommend the YouTube playlist by **Keerti Purswani**.

In my opinion, the most challenging part of the process is the OA because you need to know basic Python for the coding question. I learned the required Python syntax in 4-5 days by solving problems on HackerRank.

Overall, I would say this is a very achievable company to crack. Even though I was unprepared after my internship at Barclays, I was able to cover all the necessary topics in about a week. Another plus is that they never ask about your projects.